F7AB - Actuarial and Financial Mathematics
Tutorial 9 – Duration, Convexity and Immunisation

Question 1 Assume a constant force of interest of 6% p.a. and time is measured in years.

1. Calculate the Macaulay duration of the following sets of cashflows.

   Set A: £300 p.a. paid continuously at a constant rate from $t = 0$ to $t = 10$.

   Set B: £300 p.a. paid continuously at a constant rate from $t = 0$ to $t = 10$ and a discrete payment of £400 at $t = 3$.

   Set C: a continuous payment paid from $t = 0$ to $t = 10$. The initial rate of payment of £300 p.a, and increases continuously thereafter at the rate 5% p.a. effective. Thus the rate of payment is $\rho(t) = 300(1.05)^t$, for $t \in [0, 10]$.

2. Using the calculated duration of Set A, explain why Set B has a lower duration than Set A. Does this depend on the chosen force of interest?

Question 2 An insurer has a liability to pay £150 million 8 years from now. The current market force of interest is 6% p.a. The insurer has available for investment a zero-coupon bond (ZCB) that matures in 5 years time and a ZCB that matures in 15 years' time.

1. Calculate the net present value, duration and convexity of the liabilities.

2. Calculate the nominal amounts of the ZCBs that need to be bought to ensure that the insurer is immunized against small changes in the force of interest.

3. By discounting the nominal amounts at the market force of interest, calculate the monetary amounts to be invested in each ZCB, i.e. how much money the insurer needs to spend now in order to buy the required nominal amounts of the ZCBs.

4. Verify that the convexity condition has been satisfied.

Question 3 Assuming that $NPV_A > 0$, prove that the three conditions in Redington's theorem are equivalent to:

- $NPV_A = NPV_L$,
- $NPV'_A = NPV'_L$, and
- $NPV''_A > NPV''_L$.

Question 4 An insurer has a liability to pay £10 million p.a. at the end of each of the next 15 years, i.e. at times $t = 1, \ldots, 15$. The current market force of interest is 4% p.a.

The insurer has available for investment a zero-coupon bond (ZCB) that matures in 2 years' time and a ZCB that matures in 20 years' time.

1. Using the conditions in Question 3, calculate the monetary amounts that need to be invested in the ZCBs to ensure that the insurer is immunized against small changes in the force of interest.

2. By accumulating the monetary amounts at the market force of interest, calculate the nominal amounts to be bought of each ZCB.

3. Verify that the convexity condition $NPV_A'' > NPV_L''$ has been satisfied. Note: you are advised to use a spreadsheet to calculate NPV_L''.

Question 5 An insurer has a liability to pay £5 million p.a. continuously at a constant rate for the next 12 years. The current market force of interest is 4% p.a. The insurer has available for investment a zero-coupon bond (ZCB) that matures in 15 years' time and a bank account, with which the insurer can borrow or deposit money at the market interest rate.

1. Explain why the duration of a sum of money deposited in the bank account is zero years.

2. Using the conditions in Question 3, calculate the monetary amounts that need to be invested in the 15-year ZCB and the bank account to ensure that the insurer is immunized against small changes in the force of interest.

3. By accumulating the monetary amount at the market force of interest, calculate the nominal amount to be bought of the 15-year ZCB.

4. Verify that the convexity condition has been satisfied. Hint: to calculate NPV_L'', apply integration-by-parts once and then note that $(\bar{I}\bar{a})_{\overline{n}|} = \int_0^n tv^t dt$.

Question 6 An investor has to pay a lump sum of £20 000 at the end of fifteen years from now and an annuity certain of £5 000 p.a. payable half-yearly in advance for twenty-five years, starting in ten years' time. The investor currently holds an amount of cash equal to the present value of these two liabilities calculated at the market rate of interest. The investor wishes to immunize her fund against small movements in the rate of interest by investing the cash in two ZCBs, Bond X and Bond Y. The investor has decided to invest an amount in Bond X sufficient to provide a capital sum of £25 000 when Bond X is redeemed in ten years' time. The remainder of the cash is invested in Bond Y.

In order to immunize her holdings at a market rate of interest of 7% p.a. effective:

1. Calculate the amount of money invested in Bond Y, and

2. Determine the term needed for Bond Y and the redemption amount payable on the maturity date.

3. Check if the investor is immunised by checking if the convexity condition $NPV_A'' > NPV_L''$ holds. Note: you are advised to use a spreadsheet to calculate NPV_L''. Can you explain your answer? What do you advise the investor to do?

Question 7 An investor borrows £100 000 to invest in a new company. Loans from the bank are subject to a force of interest of 8% p.a. Deposits in the bank earn interest at the rate of 5% p.a. effective.

The investment will produce the following cashflows only: Nothing during the first year, followed by a continuous payment stream of £15 000 per annum from time 1 to time 15, and finally a lump sum of £10 000 at time 15.

1. Calculate the discounted payback period on the loan.

2. Calculate the accumulated profit at time 15.

Answers: Q1.1. $\tau_A = 4.503$ years, $\tau_B = 4.309$ years, $\tau_C = 4.907$ years; Q2.1. $\text{NPV}_L = £92.818\text{m}$, $\tau_L = 8$ years, $CONV_L = 64$ years^2; Q2.2. £87.704m of 5-year ZCB, £68.489m of 15-year ZCB; Q2.3. £64.973m of 5-year ZCB, £27.846m of 15-year ZCB; Q4.1. £78.263m of 2-year ZCB, £32.293m of 20-year ZCB; Q4.2 £84.781m of 2-year ZCB, £71.869m of 20-year ZCB; Q5.2. £17.542m of 15-year ZCB, £30.110m in bank account; Q5.3. £31.964m; Q6.1. £25 706.99; Q6.2. Term 22.182 years, redemption proceeds £115 303.42; Q7.1. 11.777 years; Q7.2. £62 353.47.